

28 Supersymmetric Versions of the Standard Model

28.2 Supersymmetry and Strong–Electroweak Unification

We shall have to defer a detailed assessment of supersymmetric models of particle physics until we are ready to consider how supersymmetry is broken. In this section we will consider the quantitative application of supersymmetry in one context in which the mechanism for the breakdown of supersymmetry is relatively unimportant, and in which supersymmetry has scored what so far is its greatest empirical success.

If the $SU(3) \cdot SU(2) \cdot U(1)$ gauge group of the strong and electroweak interactions is embedded in a simple group G that has the known quarks and leptons (plus perhaps some $SU(3) \cdot SU(2) \cdot U(1)$ -neutral fermions) as a representation, then, as described in Section 21.5, at energies at or above the scale M_X at which G is spontaneously broken, the $SU(3) \cdot SU(2) \cdot U(1)$ coupling constants will be related by

$$g_s^2 = g^2 = \frac{5g'^2}{3} \quad \text{at energies} \geq M_X. \quad (28.2.1)$$

At energies far below M_X , these couplings are seriously affected by renormalization corrections. If measured at a scale $\mu < M_X$, the couplings will have values $g'^2(\mu)$, $g^2(\mu)$, $g_s^2(\mu)$, governed by one-loop renormalization group equations

$$\mu \frac{d}{d\mu} g'(\mu) = \beta_1(g'(\mu)), \quad \mu \frac{d}{d\mu} g(\mu) = \beta_2(g(\mu)), \quad \mu \frac{d}{d\mu} g_s(\mu) = \beta_3(g_s(\mu)), \quad (28.2.2)$$

with initial conditions at M_X satisfying Eq. (28.2.1). In the original use [5] of these renormalization group equations, discussed in Section 21.5, the beta functions were calculated in one-loop order to be

$$\beta_1 = \frac{5n_g g'^3}{36\pi^2}, \quad (28.2.3)$$

$$\beta_2 = \frac{g^3}{4\pi^2} \left(-\frac{11}{6} + \frac{n_g}{3} \right), \quad (28.2.4)$$

$$\beta_3 = \frac{g_s^3}{4\pi^2} \left(-\frac{11}{4} + \frac{n_g}{3} \right), \quad (28.2.5)$$

where n_g is the number of generations of quarks and leptons and the relatively small contributions of scalar fields are here neglected. Since M_X will turn out to be many orders of magnitude larger than the energies accessible with today's accelerators, it seems reasonable to suppose that supersymmetry is unbroken over most of the range below M_X , in which case all of the new fields discussed in the previous section need to be included in calculations of the beta functions in Eq. (28.2.1). These new fields introduce three major changes in the calculations of the beta functions:

1. For every gauge boson, there is a Majorana gaugino with the same $SU(3) \cdot SU(2) \cdot U(1)$ quantum numbers. Equation (17.5.41) shows that the ratio of the contribution to the beta function for any gauge coupling of a Dirac fermion that furnishes a representation of the gauge group with generators t_A to the contribution of the corresponding gauge boson is

$-4C_2/11C_1$, where according to Eqs. (17.5.33) and (17.5.34) the ratio of C_1 and C_2 is given by:

$$\sum_{AB} C_{CAB} C_{DBA} = -(C_1/C_2) \text{Tr}(t_C t_D). \quad (28.2.6)$$

For the adjoint representation, $(t_C)_{AB} = iC_{ABC}$, so $C_1 = C_2$, and so a Dirac fermion in the adjoint representation makes a contribution that is $-4/11$ that of the gauge bosons. But the gauginos are Majorana fermions, so their contribution is $-2/11$ that of the gauge boson. Thus the term $11/6$ and $11/4$ in Eqs. (28.2.4) and (28.2.5) are reduced by a factor $9/11$ to $9/6$ and $9/4$, respectively.

2. For every left-handed quark, lepton, antiquark, or antilepton field, there is a complex scalar field with the same $SU(3) \cdot SU(2) \cdot U(1)$ quantum numbers. Following the same method as in Section 17.5, it is not hard to calculate that the contribution of a complex scalar field belonging to a representation of a gauge group with generators t_A to the beta function for the gauge coupling g_i is

$$[\beta_i(g_i)]_{\text{scalar}} = \frac{g_i^3 C_{2i}}{48\pi^2}, \quad (28.2.7)$$

where $\text{Tr}(t_A t_B) = g_i^2 C_{2i} \delta_{AB}$. This is $1/4$ the contribution of a Dirac spinor field in the same representation, given by Eq. (18.7.2), and hence $1/2$ the contribution of each left-handed spinor field (including the complex conjugates of the right-handed components of the Dirac fields). Thus the coefficients of n_g in Eqs. (28.2.3)–(28.2.5) should be increased by a factor $3/2$.

3. The decrease by a factor $9/11$ of the negative gauge boson terms in the beta function and the increase by a factor $3/2$ of the positive squark and slepton terms both lead to a general decrease in the rate at which the three gauge coupling constants diverge below M_X from the ratios (28.2.1). This will increase our estimate of M_X , but, as we shall see, in itself it would have no affect on the prediction of the electroweak mixing parameter $\sin^2 \theta$. But these changes do enhance the relative contribution of the Higgs scalars, which was neglected in Eqs. (28.2.3)–(28.2.5), and which is now also accompanied with the larger contribution of the accompanying higgsinos. With n_s of the superfields (H_1^0, H_1^-) or (H_2^+, H_2^0) discussed in the previous section, the constant C_{2i} in Eq. (28.2.7) is $[(1/2)^2 + (-1/2)^2]n_s = n_s/2$ for $SU(2)$, and is $2n_s(\pm 1/2)^2 = n_s/2$ for $U(1)$. According to Eq. (28.2.7), the scalar components of these superfields make a contribution to β_1 equal to $n_s g'^3/96\pi^2$ and a contribution to β_2 also equal to $n_s g^3/96\pi^2$. As we have seen, the Majorana higgsinos contribute twice as much to the beta functions as complex scalars with the same quantum numbers, so the superfields (H_1^0, H_1^-) or (H_2^+, H_2^0) make a total contribution to β_1 and β_2 that is $3/2$ the contribution of the Higgs scalars, and hence equal to $n_s g'^3/32\pi^2$ and $n_s g^3/32\pi^2$, respectively.

Making all these changes in the beta functions, we now have

$$\beta_1 = \frac{g'^3}{4\pi^2} \left(\frac{5n_g}{6} + \frac{n_s}{8} \right), \quad (28.2.8)$$

$$\beta_2 = \frac{g^3}{4\pi^2} \left(-\frac{9}{6} + \frac{n_g}{2} + \frac{n_s}{8} \right), \quad (28.2.9)$$

$$\beta_3 = \frac{g_s^3}{4\pi^2} \left(-\frac{9}{4} + \frac{n_g}{2} \right). \quad (28.2.10)$$

The solutions of the renormalization group equations (28.2.2) are then

$$\frac{1}{g'^2(\mu)} = \frac{1}{g'^2(M_X)} + \frac{1}{2\pi^2} \left(\frac{5n_g}{6} + \frac{n_s}{8} \right) \ln \left(\frac{M_X}{\mu} \right), \quad (28.2.11)$$

$$\frac{1}{g^2(\mu)} = \frac{1}{g^2(M_X)} + \frac{1}{2\pi^2} \left(-\frac{3}{2} + \frac{n_g}{2} + \frac{n_s}{8} \right) \ln \left(\frac{M_X}{\mu} \right), \quad (28.2.12)$$

$$\frac{1}{g_s^2(\mu)} = \frac{1}{g_s^2(M_X)} + \frac{1}{2\pi^2} \left(-\frac{9}{4} + \frac{n_g}{2} \right) \ln \left(\frac{M_X}{\mu} \right). \quad (28.2.13)$$

It is convenient to take $\mu = m_Z$, so that $SU(2) \cdot U(1)$ can be regarded as unbroken over almost all of the range of energies in which we use the formulas (28.2.11)–(28.2.13). Using Eq. (28.2.1), the difference between Eqs. (28.2.12) and (28.2.13) gives

$$\frac{1}{g^2(m_Z)} - \frac{1}{g_s^2(m_Z)} = \frac{1}{2\pi^2} \left(\frac{3}{4} + \frac{n_s}{8} \right) \ln \left(\frac{M_X}{m_Z} \right), \quad (28.2.14)$$

Table 28.1. Values of the electroweak mixing parameter $\sin^2 \theta$ and unification mass M_X given by Eqs. (28.2.17) and (28.2.18), as functions of the number n_s of left-chiral superfield doublets (H_1^0, H_1^-) or (H_2^+, H_2^0) .

n_s	$\sin^2 \theta$	M_X (GeV)
0	0.203	$8.7 \cdot 10^{17}$
2	0.231	$2.2 \cdot 10^{16}$
4	0.253	$1.1 \cdot 10^{15}$

while the difference between Eq. (28.2.12) and 3/5 of Eq. (28.2.11) gives

$$\frac{1}{g^2(m_Z)} - \frac{3}{5g'^2(m_Z)} = \frac{1}{2\pi^2} \left(-\frac{3}{2} + \frac{n_s}{20} \right) \ln \left(\frac{M_X}{m_Z} \right). \quad (28.2.15)$$

Equation (21.3.19) allows us to express the electroweak couplings in terms of the electroweak mixing angle θ and the positron charge e :

$$g(m_Z) = -e(m_Z)/\sin \theta, \quad g'(m_Z) = -e(m_Z)/\cos \theta. \quad (28.2.16)$$

We can then solve for the unknowns $\ln(M_X/m_Z)$ and $\sin^2 \theta$ in terms of input parameters $e(m_Z)$ and $g_s(m_Z)$:

$$\sin^2 \theta = \frac{18 + 3n_s + (e^2(m_Z)/g_s^2(m_Z))(60 - 2n_s)}{108 + 6n_s}, \quad (28.2.17)$$

$$\ln \left(\frac{M_X}{m_Z} \right) = \left(\frac{8\pi^2}{e^2(m_Z)} \right) \left(\frac{1 - (8e^2(m_Z)/3g_s^2(m_Z))}{18 + n_s} \right). \quad (28.2.18)$$

For $n_s = 0$ Eq. (28.2.17) gives the same result (21.5.15) for $\sin^2 \theta$ as was originally calculated (ignoring the small contribution of Higgs scalars) in non-supersymmetric theories, but the value (28.2.18) of $\ln(M_X/m_Z)$ is larger than the original result (21.5.16) by a factor 11/9, which as we have seen arises from gaugino contributions to the beta functions.

Using the same input parameters $e^2(m_Z)/4\pi = (128)^{-1}$, $g_s^2(m_Z)/4\pi = 0.118$, $m_Z = 91.19$ GeV as in Section 21.5 now gives the numerical results shown in Table 28.1. As discussed in the previous section, the necessity of cancelling anomalies in the electroweak currents requires equal numbers of (H_1^0, H_1^-) and (H_2^+, H_2^0) doublets, so we consider only even values of the number n_s of these superfields.

Remarkably, the value $n_s = 2$ for the simplest plausible theory yields a value [6] $\sin^2 \theta = 0.231$ which is in perfect agreement with the experimentally observed value, $\sin^2 \theta = 0.23$. The value of M_X is 20 times greater [7] than calculated in this way in non-supersymmetric theories, leading to a decrease by a factor 20^{-4} in the rate for proton decay processes like $p \rightarrow \pi^0 + e^+$, thus removing a conflict with the experimental non-observation of such processes. (Proton decay is discussed in more detail in Section 28.7.) This increase in the value of M_X brings it closer to the energy scale $\approx 10^{18}$ GeV at which gravitation has the same strength as other interactions. It may be that this remaining gap may be filled by a change in gravitational interactions at very high energies [7a].

A value $n_s = 4$ would give a value for $\sin^2 \theta$ in serious disagreement with experiment, and a value of M_X low enough to revive the conflict with expectations for proton decay. This makes a strong case for having just one of each superfield (H_1^0, H_1^-) and (H_2^+, H_2^0) .

Unlike the calculated values of $\sin^2 \theta$ and M_X , the calculated value of the common gauge coupling (28.2.1) at M_X does depend on the number of generations as well as the number of scalar doublets. With $n_g = 3$ and $n_s = 2$ and our previous input parameters, Eq. (28.2.13) gives

$$\frac{g^2(M_X)}{4\pi} = \frac{g_s^2(M_X)}{4\pi} = \frac{1}{17.5}. \quad (28.2.19)$$

28.3 Where is Supersymmetry Broken?

Supersymmetry if valid at all is certainly not apparent in the menu of known particles, so any consideration of the implications of supersymmetry at ordinary energies requires us to make some assumption about the mechanism of supersymmetry breaking. It would be simplest to suppose that supersymmetry is broken like $SU(2) \cdot U(1)$, by effects occurring in the tree approximation of the supersymmetric standard model. This possibility may be definitely ruled out.

One argument against a tree-approximation breakdown of supersymmetry is based on the mass sum rule (27.5.11), which holds separately for each value of the unbroken conserved quantities color and electric charge. In the color-triplet sector with electric charge $-e/3$ the only known fermions are the d , s , and b quarks, for which

$$m_d^2 + m_s^2 + m_b^2 \simeq (5 \text{ GeV})^2. \quad (28.3.1)$$

According to the sum rule, if there are no other fermions with this color and charge, then the sum of all squared masses for bosons (counting each spin state separately) with the

same color and charge must equal about $2(5 \text{ GeV})^2$. In particular, each of the squarks with this color and charge must have a mass no greater than 7 GeV. The existence of such light squarks is definitely ruled out experimentally; they would have shown up, for instance, as a contribution to the rate for electron–positron annihilation into hadrons at energies where this process has been studied very thoroughly.

This argument could be invalidated if there were a heavy fourth generation of quarks. There is another argument, due to Dimopoulos and Georgi [3], which would apply however many heavy quarks there are and which yields an even stronger upper bound on the mass of the lightest squark. The unbroken conservation of charge and color tells us that the only non-zero D_{A0} -terms in the supersymmetric standard model are for the generators y of $U(1)$ and t_3 of $SU(2)$, which we shall call D_1 and D_2 , respectively. The values of these generators are $y = -g'/6$ and $t_3 = +g/2$ for the left-handed quarks of charge $2e/3$; $y = -g'/6$ and $t_3 = -g/2$ for the left-handed quarks of charge $-e/3$; $y = 2g'/3$ and $t_3 = 0$ for the right-handed quarks of charge $2e/3$; and $y = -g'/3$ and $t_3 = 0$ for the right-handed quarks of charge $-e/3$. Also, the squark fields are color-triplets, and therefore cannot have vacuum expectation values. According to Eq. (27.5.4), the mass-squared matrix of the charge $2e/3$ color-triplet (not antitriplet) squarks is

$$M_{0U}^2 = \begin{bmatrix} \mathcal{M}_U^* \mathcal{M}_U - g' D_1/6 + g D_2/2 & \mathcal{F}_U^* \\ \mathcal{F}_U & \mathcal{M}_U \mathcal{M}_U^* + 2g' D_1/3 \end{bmatrix}, \quad (28.3.2)$$

while the mass-squared matrix of the charge $-e/3$ color-triplet squarks is

$$M_{0D}^2 = \begin{bmatrix} \mathcal{M}_D^* \mathcal{M}_D - g' D_1/6 - g D_2/2 & \mathcal{F}_D^* \\ \mathcal{F}_D & \mathcal{M}_D \mathcal{M}_D^* - g' D_1/3 \end{bmatrix}. \quad (28.3.3)$$

Also, Eq. (27.5.6) gives the mass-squared matrices of the quarks of charge $2e/3$ and $-e/3$ here as just $\mathcal{M}_U^* \mathcal{M}_U$ and $\mathcal{M}_D^* \mathcal{M}_D$, respectively, with no mixing with the gauginos.

Now let v_u and v_d be the normalized eigenvectors of the quark mass-squared matrices $\mathcal{M}_U^* \mathcal{M}_U$ and $\mathcal{M}_D^* \mathcal{M}_D$ corresponding to the quarks u and d of lowest mass, and consider the expectation values of the corresponding squark mass-squared matrices

$$\begin{bmatrix} 0 \\ v_u^* \end{bmatrix}^\dagger M_{0U}^2 \begin{bmatrix} 0 \\ v_u^* \end{bmatrix} = m_u^2 + \frac{2g' D_1}{3}, \quad (28.3.4)$$

$$\begin{bmatrix} 0 \\ v_d^* \end{bmatrix}^\dagger M_{0D}^2 \begin{bmatrix} 0 \\ v_d^* \end{bmatrix} = m_d^2 - \frac{g' D_1}{3}. \quad (28.3.5)$$

These expectation values are weighted averages of the squared masses of the squarks of charge $2e/3$ and $-e/3$, respectively, so at least one squark of charge $2e/3$ must have a squared mass less than $m_u^2 + 2g' D_1/3$, and at least one squark of charge $-e/3$ must have a squared mass less than $m_d^2 - g' D_1/3$. Thus, depending on the sign of D_1 , *there must be either a squark of charge $2e/3$ lighter than the u quark, or a squark of charge $-e/3$ lighter than the d quark.*

Needless to say, the existence of a charged color-triplet scalar this light would radically change strong-interaction phenomenology. Like the u and d quarks, this colored scalar would appear as an ingredient of hadrons with a ‘constituent’ mass of a few hundred MeV, which is certainly not seen. Since this scalar is electrically charged, it would also be created in pairs in $e^+ - e^-$ annihilation at energies above a few hundred MeV, making a contribution to the annihilation cross-section that would destroy the excellent agreement between theory and experiment for this cross-section. Even worse, since the u and d quarks are so light, and D_1 is expected to be of the order of the supersymmetry-breaking scale, Eqs. (28.3.4) and (28.3.5) indicate that one of the squarks would have a negative squared mass, meaning that this squark field would have to develop a non-vanishing expectation value, breaking both color and charge conservation. We are forced to reject the simple picture of supersymmetry broken spontaneously in the tree approximation in a supersymmetric version of the standard model.

One way out of this conclusion would be to add another $U(1)$ gauge superfield to the theory. If all the quark superfields carry the same value $\tilde{g}$ of this new $U(1)$ generator, then the corresponding D -term $\tilde{D}$ would make an additive contribution $\tilde{g}\tilde{D}$ to the right-hand sides of both Eqs. (28.3.4) and (28.3.5). If this term were sufficiently large, then it could give a large positive value to all of the squark squared masses, avoiding all the problems mentioned above. But there is no sign of such a new neutral gauge boson at accessible energies, and in any case we would still have an upper bound of 7 GeV on the masses of all the squarks of charge $-e/3$.

It is not necessarily a bad thing that we have to look for the breaking of supersymmetry elsewhere than in the tree approximation of the supersymmetric standard model. If supersymmetry were broken in this approximation, then the characteristic mass that sets the scale of supersymmetry breaking would be some mass parameter in the Lagrangian, which would in turn set the scale of all other masses in the standard model. We would then still be confronted with the hierarchy problem: why is this mass scale so much less than 10^{16} – 10^{18} GeV?

There is one known way to explain such large mass ratios. If supersymmetry is not spontaneously broken in the tree approximation in whatever field theory unifies all the interactions at some high mass scale M_X , then as shown in Section 27.6, it will not be broken in any order of perturbation theory. But it can be broken by non-perturbative effects. In particular, if there is some gauge field with an asymptotically free gauge coupling $\mathcal{G}(\mu)$ at renormalization scale μ , and if $\mathcal{G}^2(\mu)/8\pi^2$ is substantially less than unity for $\mu \approx M_X$, then as discussed in Section 18.3, this gauge interaction will become strong at an energy of order $M_S = M_X \exp(-8\pi^2 b/\mathcal{G}^2(M_X))$, where b is a number of order unity. It is not necessary for $\mathcal{G}^2(M_X)/8\pi^2$ to be very small in order to have M_S many orders of magnitude less than M_X . We will see in Section 29.4 that supersymmetry can indeed be broken in just this way, by a gauge coupling that has become strong at some energy $M_S \ll M_X$. Indeed, this is just what happens to chiral symmetry in quantum chromodynamics; there is no mystery why the proton mass (or at least its main part, due to the dynamical breakdown of chiral symmetry, and not to the tiny masses of the u and d quarks) is so much smaller than the unification scale M_X . Alternatively, the forces that are strong at energy M_S may produce

a potential for scalar fields, whose vacuum expectation value then breaks supersymmetry.

There are no signs of any new strong interaction of the known quarks and leptons, so we have to assume that the observed particles of the standard model are neutral with respect to the strong force that breaks supersymmetry. Supersymmetry breaking therefore occurs in a ‘hidden sector’ of particles that do feel this new strong force. The remaining question, then, is what is the mechanism by which supersymmetry breaking in this hidden sector is communicated to the known particles of the standard model? As we shall see, most of our expectations for the phenomenological implications of supersymmetry depend on the answer to this question, rather than on the details of the breakdown of supersymmetry itself.

Of course, the mechanism for communicating supersymmetry breaking to observed particles must be some sort of interaction that is felt by these particles. There are two leading candidates. One mechanism is provided by the $SU(3) \cdot SU(2) \cdot U(1)$ gauge interactions themselves, to be discussed in Section 28.6. The other is gravitation, or rather the auxiliary fields that are superpartners of the gravitational field, to be discussed in Sections 31.4 and 31.7.

Without going into details here, we can make a crude estimate of the supersymmetry-breaking scale M_S for these two possibilities. For gauge-mediated supersymmetry breaking, we expect that the mass splitting between the observed quarks, leptons, and gauge bosons and their superpartners would be of order $g_s^2/16\pi^2$ or $g'^2/16\pi^2$ or $g^2/16\pi^2$ (where g_s , g , and g' are the $SU(3)$, $SU(2)$, and $U(1)$ gauge couplings), depending on which quantum numbers are carried by the supermultiplet in question. (This guess is verified in Section 28.6.) Hence if the squarks, sleptons, and gauginos have masses in the range of 100 GeV to 10 TeV, as argued at the end of Section 28.1, then the supersymmetry-breaking scale M_S would be higher by two or three orders of magnitude—say, of order 100 TeV. On the other hand, if it is gravitation that serves as the mediator of supersymmetry breaking, then on dimensional grounds we would expect the mass splittings Δm between the observed particles and their superpartners to be of order $\sqrt{G}M_S^2$, or perhaps of order GM_S^3 . (Results of both sorts will be encountered in models described in Section 31.7.) If the squarks, sleptons, and gauginos have masses in the range of 100 GeV to 10 TeV, then M_S would be of order 10^{11} GeV for $\Delta m \approx \sqrt{G}M_S^2$, or 10^{13} GeV for $\Delta m \approx GM_S^3$.

The large difference in estimates of the supersymmetry-breaking scale M_S for gauge- and gravitation-mediated supersymmetry breaking makes an important difference in particle phenomenology and cosmology. As already mentioned several times, supersymmetry dictates that the graviton must have a partner of spin 3/2, the gravitino. When supersymmetry is spontaneously broken at a scale M_S , the gravitino acquires a mass m_g of order $\sqrt{G}M_S^2$. (A precise formula will be given in Section 31.3.) For gauge-mediated supersymmetry breaking, this is very small; if $M_S \approx 100$ TeV then $m_g \approx 1$ eV, so the gravitino would be by far the lightest of the new particles required by supersymmetry—that is, the lightest particle with negative R parity (28.1.6). On the other hand, for gravitationally mediated supersymmetry breaking the gravitino mass is just of the same order of magnitude $\sqrt{G}M_S^2$ as the mass splitting between known particles and their superpartners, so the gravitino would have roughly the same mass as the squarks, sleptons, and gauginos. The gravitino

then might or might not be the lightest particle with negative R parity, but its interactions with known particles and their superpartners in this case are of gravitational strength, so that gravitinos would play no direct role in experiments on elementary particles.

* * *

There are limits on the number of gravitinos that could survive from the big bang, which set useful constraints on the scale M_S of supersymmetry breaking. At some point in the distant past the temperature T was presumably high enough so that even purely gravitational interactions would have kept gravitinos in thermal equilibrium with other particles, in which case the number density of gravitinos would have been of the order of T^3 , roughly the same as the number density of photons. (We are using units in which the Boltzmann constant k_B as well as $\hbar$ and c are equal to unity.) If gravitinos do not annihilate or decay, then the expansion of the universe will lower their number density in the same way as the number density of photons, so even after the gravitinos go out of equilibrium they would be present in numbers comparable to those of photons. More precisely, since the photons but not the gravitinos are heated by the annihilation of other particles, the number density n_{g0} of gravitinos at present would be one or two orders of magnitude less than the number density $n_{\gamma 0}$ of photons in the cosmic microwave radiation background. In order for the mass density $m_g n_{g0}$ of the gravitinos not to exceed the upper bound on the cosmic mass density set by the observed value of the Hubble constant, m_g would have to be less [8] than about 1 keV. As we have seen, this limit is well satisfied in theories of gauge-mediated supersymmetry breaking, where the gravitino is too light for cosmic gravitinos to contribute appreciably to the mass density of the universe. Since some of the fields that break supersymmetry in these theories must interact at least indirectly with the known quark, lepton, and gauge fields in order for the known particles to show the effects of supersymmetry breaking, the interactions of the gravitino with the known particles and their superpartners are suppressed only by powers of gauge and Yukawa coupling constants, so all the superpartners of the quarks, leptons, and gauge bosons would decay quickly into these known particles and gravitinos. Thus these particles also do not provide candidates in these models for the ‘missing mass’ sought by cosmologists. (It is possible that conservation laws could keep some particles of the supersymmetry-breaking sector stable, in which case they could conceivably serve as the missing mass.)

On the other hand, for supersymmetry breaking that is gravitationally mediated, gravitinos are heavy enough to be unstable (though gravitino annihilation is still negligible), so the above limit need not apply [9]. We will see in Section 31.3 that the coupling of the gravitino to other fields is proportional to $\sqrt{G}$, so on dimensional grounds the decay rate Γ_g of a gravitino at rest is roughly of the order of Gm_g^3 . This is to be compared with the rate of expansion of the universe, which at temperature T is of order $\sqrt{GT^4}$. (We are here ignoring factors of order 10–100, including those involving non-gravitational coupling constants and the number of particle species.) When the cosmic temperature drops to the value $T \approx m_g$ at which gravitinos become non-relativistic, the ratio of their decay rate to the expansion rate is of order $\sqrt{G}m_g = m_g/m_{\text{Planck}} \ll 1$, so gravitino decay becomes significant

only after this time, when the gravitinos are highly non-relativistic. As we have seen, their number density will be of order T^3 , so their energy density will then be of order $m_g T^3$, which is greater than the energy density of order T^4 of the photons and other particles in thermal equilibrium at temperature T , and therefore makes the dominant contribution to the cosmic gravitational field that governs the rate of expansion of the universe. The expansion rate under these conditions is therefore of order $\sqrt{G m_g T^3}$, and gravitino decay becomes significant when this equals the gravitino decay rate of order $G m_g^3$, and therefore at a temperature

$$T_g \approx G^{1/3} m_g^{5/3}.$$

As we have seen, if these gravitinos did not decay before the present then their mass had better be less than 1 keV, but they can lead to cosmological difficulties even if they did decay before now. After they decay, their energy must go into the energy of photons and other relativistic particles, so the temperature T'_g after decay is related to the temperature T_g calculated above by the energy conservation condition $m_g T_g^3 \approx T'^4_g$, and hence

$$T'_g \approx G^{1/4} m_g^{3/2}.$$

In particular, since $T_g \ll m_g$, we have $T'_g \gg T_g$. If T_g were less than the temperature $T_n \simeq 0.1$ MeV at which cosmological nucleosynthesis can occur, then gravitinos would still be abundant before nucleosynthesis, giving a higher energy density and hence a faster expansion, so that there would be less time for free neutrons to decay before being incorporated into complex nuclei, and hence more helium would be produced when nucleosynthesis occurs. Also, the ratio of the photon and baryon densities would have been subsequently increased by gravitino decay, so this ratio at the time of nucleosynthesis would have been considerably less than is usually estimated from the present cosmic microwave background temperature, and so nuclear reactions would have incorporated neutrons more completely into helium, and less deuterium would be left today. The present agreement between theory and observation for the cosmic helium and deuterium abundances would thus be destroyed. This problem is avoided if $T_g > 0.1$ MeV, but it can also be avoided under the much weaker condition that $T'_g > 0.4$ MeV, because then after the gravitinos decay the temperature would have been high enough to break up the excess helium and give cosmological nucleosynthesis a fresh start as the universe recools. This condition requires that $m_g > 10$ TeV, which is just barely consistent with the upper bound derived in Section 28.1 on the masses of the superpartners of the known quarks, leptons, and gauge bosons, which for gravitationally mediated supersymmetry breaking are of order m_g . This limit on m_g corresponds to a supersymmetry-breaking scale $M_S > 10^{11}$ GeV for $m_g \approx \sqrt{G} M_S^2$ or $M_S > 10^{13}$ GeV for $m_g \approx G M_S^3$.